

LINEAR ALGEBRA COURSE — FINAL PROJECT REPORT

Development of a Linear Algebra-Based Foreign Trade Decision Support System for Turkey Export Prediction

Data Sources:

Kaggle DS1 — Awan, M.T. (2023). World Export & Import Dataset

<https://www.kaggle.com/datasets/muhammadtalhaawan/world-export-and-import-dataset>

Kaggle DS2 — Gencheva, V. (n.d.). Macro-Economic Indicators Dataset

<https://www.kaggle.com/datasets/veselagencheva/macro-economic-indicators-dataset-country-level>

Kaggle DS3 — Appetukhov, A. (2022). International Trade Database

<https://www.kaggle.com/datasets/appetukhov/international-trade-database>

Methods: Normal Equation (LSQ) · SVD Pseudo-Inverse · PCA Regression

TABLE OF CONTENTS

ABSTRACT	3
1. INTRODUCTION	4
1.1 General Assessment and Problem Definition	4
1.2 Historical Development of Turkey's Foreign Trade	4
1.3 Literature Review	5
1.4 Contribution and Scope of the Study	6
2. METHODOLOGY	7
2.1 Dataset, Sources and Variables	7
2.2 Cross-Validation Method	7
2.3 Data Preprocessing and Standardization	8
2.4 Method 1: Normal Equation	8
2.5 Method 2: SVD Pseudo-Inverse	9
2.6 Method 3: PCA Regression	9
2.7 Performance Metrics	10
3. FINDINGS	10
3.1 Descriptive Statistics	10
3.2 Cross-Validation Results	11
3.3 SVD Singular Value Analysis	11
3.4 PCA Component Analysis	12
3.5 Regression Coefficients	12
3.6 Model Performance Comparison	13
3.7 Actual vs Predicted Values	13
3.8 PCA Coefficient Comparison	14
4. CONCLUSION AND RECOMMENDATIONS	14
REFERENCES	16

ABSTRACT

Foreign trade is one of the fundamental macroeconomic indicators that directly determines the growth dynamics of national economies. In developing economies, export growth directly affects foreign exchange revenues, employment levels, and technology transfers.

This study comparatively applies Normal Equation (Least Squares), Singular Value Decomposition (SVD) pseudo-inverse, and Principal Component Analysis (PCA) methods to predict Turkey's export performance for the 2000–2021 period through macroeconomic variables. Data were compiled from three open datasets obtained from the Kaggle platform and their consistency was tested through cross-validation.

Normal Equation and SVD models produced $R^2 = 0.9981$, $RMSE = 3.70$ Billion USD, $MAE = 2.96$ Billion USD; while PCA Regression produced $R^2 = 0.8657$, $RMSE = 31.21$ Billion USD. The SVD condition number $\kappa(X) = 7.85$, and the Gram matrix condition number $\text{cond}(X^T X) = 61.62$ were calculated. The Euclidean norm difference between Normal Equation and SVD solutions remained at $\|\beta_{\text{lstsq}} - \beta_{\text{svd}}\| = 3.04 \times 10^{-13}$; and the trade openness variable emerged as the strongest positive determinant with $\beta_4 = +87.56$.

It is recommended that export policy be supported by structural reforms increasing trade openness and implemented in coordination with monetary policy prioritizing inflation control.

Keywords: foreign trade forecasting, linear regression, SVD, PCA, Turkey exports

1. INTRODUCTION

1.1 General Assessment and Problem Definition

Forecasting trade flows is critically important for designing export policies, managing exchange rates, and maintaining macroeconomic stability. The fundamental research question of this study can be stated as follows: To what extent can five macroeconomic variables—country growth rate, inflation, MFN tariff rate, trade openness, and import/export ratio—explain Turkey's annual exports? To answer this question, three different linear algebra methods—Normal Equation, SVD, and PCA—are comparatively applied.

1.2 Historical Development of Turkey's Foreign Trade

The Turkish economy adopted an export-led growth strategy within the framework of liberalization policies implemented in the 1980s. The liberalization of capital movements in 1989 and the establishment of the Customs Union with the European Union in 1996 were the most decisive turning points in this process.

According to data obtained from the Kaggle dataset (Awan, 2023), Turkey's total exports increased continuously from 77.53 Billion USD in 2000 to 330.89 Billion USD in 2021. During this period, MFN tariff rates systematically declined (2000: 11.93% → 2021: 8.59%), while the trade openness index increased significantly. The 2001 economic crisis pushed exports down to 55.36 Billion USD; the 2008–2009 global financial crisis led to a sharp decline following the 2008 peak of 252 Billion USD.

This dynamic structure provides a strong testing ground for the linear model. In particular, how sudden changes during crisis years are captured by the model serves as a critical criterion for methodological evaluation.

1.3 Literature Review

Studies examining export determinants encompass a wide methodological spectrum ranging from gravity models to dynamic panel analyses. Tinbergen (1962) introduced the gravity model to economics literature, explaining trade flows between countries through economic size and geographic distance. This approach continues to serve as a fundamental reference point in trade policy research today.

Helpman and Krugman (1985) made significant contributions to this field with new trade theory emphasizing economies of scale and product differentiation. Regarding export elasticities, Goldstein and Khan (1985) comprehensively addressed the effects of exchange rate and income variables on exports within standard economic models.

Regarding econometric applications of linear algebra methods, Strang (2006) thoroughly examined the Normal Equation and SVD-based Least Squares solution. Emphasizing that

high condition numbers of the Gram matrix can lead to numerical stability problems, Strang proposed SVD as a more reliable alternative. The low-rank matrix approximation theorem proved by Eckart and Young (1936) mathematically demonstrates that the SVD solution is optimal.

Jolliffe (2002) comprehensively demonstrated the role of PCA in addressing multicollinearity problems and in dimensionality reduction for high-dimensional datasets. It serves as a fundamental reference on how the correlation structure between variables affects regression coefficients and how PCA resolves this problem.

The relationship between tariff rates and export performance continues to be an important research topic. The MFN tariff series observed in our dataset (Awan, 2023) reveals a systematic decline from high levels in the early 2000s; this trend is consistent with the WTO trade liberalization process.

1.4 Contribution and Scope of the Study

The present study makes original contributions in four fundamental respects: (1) Reproducibility is ensured by using open and verifiable Kaggle datasets. (2) Systematic cross-validation is performed between two different Kaggle sources. (3) Three linear algebra methods are comparatively evaluated on the same data. (4) Methodological findings are supported with a practical tool through an interactive HTML-based decision support interface.

2. METHODOLOGY

2.1 Dataset, Sources and Variables

Three open datasets obtained from the Kaggle platform were used in the study. The World Export & Import Dataset compiled by Awan (2023) was selected as the primary source. This dataset consists of 8,096 rows and 33 columns, covering country-level export, import, and tariff data for the years 1988–2021. The secondary source, Gencheva (n.d.) Macro-Economic Indicators Dataset (DS2), was used to verify inflation data for 2020–2021. The third source, Appetukhov (2022) International Trade Database (DS3), provided cross-validation with 634,509 rows of UN Comtrade data. The analysis period was set as 2000–2021 ($n = 22$ years).

Table 1

Variable Definitions, Units of Measurement and Kaggle Data Sources

Sym.	Variable	Unit	Source	Description
y	Export	Billion USD	DS1	Turkey total goods exports
x ₁	Country Growth Rate	%	DS1	Country Growth (%)
x ₂	Inflation (CPI)	%	DS1+DS2	DS2 cross-verified 2020–21
x ₃	MFN Tariff Rate	%	DS1	Most Favoured Nation avg. tariff
x ₄	Trade Openness	Index	DS1	(Exp+Imp) based normalised index
x ₅	Imp/Exp Ratio	Ratio	DS1	Imports divided by exports
	Note. DS1 = Awan (2023); DS2 = Gencheva (n.d.); DS3 = Appetukhov (2022). All sources from Kaggle.			

2.2 Cross-Validation Method

To test data reliability, DS1 and DS3 export series were compared. Since DS3 originates from UN Comtrade, it leads to systematic deviation due to its coverage of re-exports and transit trade flows. During the 2019–2021 period, the deviation exceeds 35%. For this reason, DS1, which measures direct goods exports, was preferred as the primary source.

2.3 Data Preprocessing and Standardization

Z-score standardization was applied for variables with different scale units: $_{ij} = (x_{ij} - \mu_j) / \sigma_j$. Standardization parameters: $\mu = [4.86, 16.01, 9.41, 100.0, 0.504]$; $\sigma = [9.54, 13.75, 1.06, 40.12, 0.071]$. The design matrix $X \in \mathbb{R}^{22 \times 6}$ was obtained by adding a constant term to the standardized matrix. The rank of the matrix was verified as $\text{rank}(X) = 6$.

2.4 Method 1: Normal Equation (Least Squares)

The least squares problem is formulated as minimizing the sum of squared residuals: $\min_{\beta} \|X\beta - y\|^2$. Closed-form solution (normal equation): $\beta^* = (X^T X)^{-1} X^T y$. Here $X^T X \in \mathbb{R}^{6 \times 6}$ is the Gram matrix and $X^T y \in \mathbb{R}^6$ is the moment vector. `numpy.linalg.solve()` was used to solve the system. The condition number of the Gram matrix was calculated as $\text{cond}(X^T X) = 61.62$ (Strang, 2006).

2.5 Method 2: SVD Pseudo-Inverse

Singular Value Decomposition: $X = U\Sigma V^T$. Here $U \in \mathbb{R}^{22 \times 6}$ is the left singular vector matrix, $\Sigma \in \mathbb{R}^{6 \times 6}$ is the diagonal singular value matrix, and $V \in \mathbb{R}^{6 \times 6}$ is the right singular vector matrix. Moore–Penrose pseudo-inverse: $X^+ = V\Sigma^+U^T \rightarrow \beta^* = X^+y$. The singular value threshold $\text{tol} = 10^{-10} \times \sigma_1$ was applied. The condition number $\kappa(X) = \sigma_1/\sigma_6 = 7.85$ is significantly lower than $\text{cond}(X^T X) = 61.62$, demonstrating the numerical stability advantage (Eckart & Young, 1936).

2.6 Method 3: PCA Regression

PCA was performed by applying SVD to the standardized matrix $\hat{X}$. Explained variance ratio: $\text{EVR}_k = \sigma_k^2 / \sum_i \sigma_i^2$. The cumulative variance threshold was set at 90%; $k = 3$ components were selected (cumulative: 92.30%). Projection: $Z = \hat{X} \cdot V_3^T \in \mathbb{R}^{22 \times 3}$. Regression was applied on Z , followed by back-projection to the original space via $\beta_{\text{pca_orig}} = V_3^T \beta_{\text{pca}}$ (Jolliffe, 2002).

2.7 Performance Metrics and Software Environment

For model evaluation, $R^2 = 1 - \Sigma(y_i - \hat{y}_i)^2 / \Sigma(y_i - \bar{y})^2$, $\text{RMSE} = \sqrt{\Sigma(y_i - \hat{y}_i)^2 / n}$ and $\text{MAE} = \Sigma|y_i - \hat{y}_i| / n$ metrics were used. All analyses were performed with Python 3.11 using NumPy 1.26 and Pandas 2.1 libraries. An HTML interface based on Chart.js 4.4 was developed for visualization.

3. FINDINGS

3.1 Descriptive Statistics

Table 2 presents the descriptive statistics of Turkey's macroeconomic variables for the 2000–2021 period. The export mean is 219.62 Billion USD with a standard deviation of 87.17 Billion USD, revealing high variance in the dataset.

Table 2

Turkey Macroeconomic Variables — Descriptive Statistics (2000–2021)

Variable	n	Mean	Std.Dev.	Min	Max	Source
Export (Billion USD)	22	219.62	87.17	55.36	330.89	DS1
Import (Billion USD)	22	112.39	51.09	25.98	213.61	DS1
Country Growth (%)	22	4.86	9.54	−16.47	18.61	DS1
Inflation (%)	22	16.01	13.75	6.30	54.90	DS1+DS2
MFN Tariff (%)	22	9.41	1.06	8.33	11.93	DS1
Trade Openness	22	100.00	40.12	25.86	164.00	DS1
Imp/Exp Ratio	22	0.504	0.071	0.416	0.645	DS1
Note. Inflation 2020–2021 values verified with DS2.						

3.2 Cross-Validation Results

Table 3 presents the comparison of DS1 and DS3 export values for selected years. Consistent results at 3.5–4.7% were obtained for 2005 and 2010, while the deviation exceeds 35% in the 2019–2021 period due to DS3's re-export coverage.

Table 3

DS1 and DS3 Export Values Cross-Validation

Year	DS1 WEI (B\$)	DS3 Intl (B\$)	Dev. (%)	Note

2005	154.13	146.95	4.7	Consistent
2010	235.93	227.77	3.5	Consistent
2015	262.01	301.96	15.2	Moderate deviation
2019	267.29	361.67	35.3	DS3 includes re-exports
2021	330.89	450.44	36.1	DS3 includes transit
Note. DS3 systematically produces higher values as it covers re-exports and transit.				

3.3 SVD Singular Value Analysis

Table 4 presents the singular values obtained from the SVD decomposition of the design matrix X . Since all singular values are nonzero, the matrix is full rank (rank = 6) and has a unique solution.

Table 4

Singular Value Decomposition of Matrix X ($X = U\Sigma V^T$)

Component	σ_i	σ_i^2	Ratio (%)	$\kappa = \sigma_1/\sigma_i$	Note
σ_1	7.6786	58.96	37.5	1.00	Largest singular value
σ_2	5.4269	29.45	18.7	1.41	
σ_3	4.6904	22.00	14.0	1.64	
σ_4	3.6226	13.12	8.3	2.12	
σ_5	2.7402	7.51	4.8	2.80	
σ_6	0.9782	0.96	0.6	7.85	$\kappa(X) = \sigma_1/\sigma_6 = 7.85$
Note. $\ \beta_{\text{lsq}} - \beta_{\text{svd}}\ = 3.04 \times 10^{-13}$. All singular values nonzero; X full rank.					

3.4 PCA Component Analysis

Table 5 presents the explained variance values of the principal component analysis. With $k = 3$ components, 92.30% of the variance was retained.

Table 5

Principal Component Analysis — Explained Variance

Component	Eigenvalue	Variance (%)	Cumulative (%)	Selection	Note
PC1	58.96	53.60	53.60	✓	Primary growth direction
PC2	29.45	26.77	80.37	✓	Tariff/inflation contrast
PC3	13.12	11.93	92.30	✓ (threshold)	Trade balance component
PC4	7.51	6.83	99.13	—	Not selected
PC5	0.96	0.87	100.00	—	Noise component
Note. $k = 3$ components selected (90% cumulative threshold exceeded).					

3.5 Regression Coefficients

Table 6 presents the standardized β coefficients obtained from Normal Equation and SVD models. The two methods are seen to produce identical coefficients.

Table 6

Normal Equation and SVD Model Standardized Regression Coefficients

Variable	β (LSQ)	β (SVD)	Direction	Interpretation
Intercept (β_0)	219.6178	219.6178	—	Mean export level (B\$)
Country Growth (β_1)	-1.8084	-1.8084	(-)	Weak negative effect
Inflation (β_2)	-3.1051	-3.1051	(-)	Cost pressure channel
MFN Tariff (β_3)	+0.8786	+0.8786	(+)	Weak positive effect
Trade Openness (β_4)	+87.5592	+87.5592	(+)	VERY STRONG positive effect
Imp/Exp Ratio (β_5)	-10.7383	-10.7383	(-)	Strong negative effect
Note. $\ \beta_{lsq} - \beta_{svd}\ = 3.04 \times 10^{-13}$. All standardized.				

3.6 Model Performance Comparison

Table 7 presents the performance metrics of the three methods comparatively. LSQ and SVD methods produce identical performance values.

Table 7

Method Performance Comparison Table

Method	R ²	RMSE (B\$)	MAE (B\$)	Variables	κ
Normal Equation (LSQ)	0.9981	3.70	2.96	6	61.62*
SVD Pseudo-Inverse	0.9981	3.70	2.96	6	7.85
PCA Regression	0.8657	31.21	25.38	3	—
Note. R ² : coefficient of determination. RMSE: root mean square error. *cond(X ^T X).					

3.7 Actual vs Predicted Values

Table 8 compares LSQ/SVD and PCA predictions with actual values for selected years.

Table 8

Actual and Model Predictions for Selected Years (Billion USD)

Year	Actual (B\$)	LSQ/SVD (B\$)	PCA (B\$)	Error %	Key Event
2000	77.53	86.02	10.07	10.9	Pre-crisis year
2001	55.36	46.93	43.08	15.2	Economic crisis
2004	134.52	135.93	195.86	1.0	Strong growth
2008	252.13	250.90	251.91	0.5	Pre-crisis peak
2009	184.07	188.58	217.85	2.4	Global crisis
2013	312.23	307.84	264.35	1.4	Trade peak
2019	267.29	270.90	273.86	1.4	
2021	330.89	333.14	308.83	0.7	Post-COVID
Note. LSQ and SVD models produce identical predictions.					

3.8 PCA Coefficient Comparison

Table 9 compares the back-projected coefficients of the PCA model to original space with LSQ coefficients. Coefficient divergence due to information loss from dimensionality reduction is observed.

Table 9

PCA Model Coefficients Back-Projected to Original Space

Variable	LSQ β	PCA β	Diff	Explanation
Country Growth	-1.8084	+8.9765	10.79	Component transformation effect
Inflation	-3.1051	-21.2138	18.11	PC1 loading strong
MFN Tariff	+0.8786	-29.4697	30.35	Largest deviation
Trade Openness	+87.5592	+31.5448	56.01	Information loss at k=3
Imp/Exp Ratio	-10.7383	+15.1363	25.88	PC2 loading effect
Note. Information loss occurs as PCA model uses k=3 components (Jolliffe, 2002).				

4. CONCLUSION AND RECOMMENDATIONS

4.1 Key Findings

4.1.1 Model Performance

LSQ and SVD methods demonstrated an extremely high fit with $R^2 = 0.9981$ and $RMSE = 3.70$ Billion USD. This value indicates that the five selected variables explain exports almost completely. The strong linear relationship between the trade openness index and exports is the primary source of this result. The PCA model produces reasonable performance with $R^2 = 0.8657$, but the information loss caused by dimensionality reduction leads to performance degradation.

4.1.2 Methodological Consistency

The value $\|\beta_{\text{lstsq}} - \beta_{\text{svd}}\| = 3.04 \times 10^{-13}$ proves that the two methods produce numerically identical results. This finding is fully consistent with the theoretical predictions of Strang (2006) and Eckart & Young (1936). SVD's solution without directly inverting the Gram matrix demonstrates its numerical superiority as the difference between $\kappa(X) = 7.85$ and $\text{cond}(X^T X) = 61.62$.

4.1.3 Dominant Role of Trade Openness

The coefficient $\beta_4 = +87.56$ reveals the dominant positive effect of the trade openness index on exports. This finding is consistent with the theoretical predictions of Helpman and Krugman (1985) regarding the positive relationship between trade liberalization and export growth. Specifically for Turkey, the European Union Customs Union emerges as the fundamental mechanism strengthening this effect.

4.1.4 Cross-Validation Finding

The primary source of systematic deviation between DS1 and DS3 (35%+ during the 2019–2021 period) is DS3's coverage of re-exports and transit trade flows within the UN Comtrade methodology framework. This methodological difference demonstrates the critical importance of source selection in interpreting international trade statistics. The data quality concerns highlighted by Goldstein and Khan (1985) are consistent with this finding.

4.2 Policy Recommendations

1. Preserving and expanding trade openness: The coefficient $\beta_4 = +87.56$ confirms the critical role of multilateral trade agreements in export growth. New trade partnerships and deepening existing agreements should be among priority policy objectives.

2. Monetary policy prioritizing inflation control: $\beta_2 = -3.11$ reveals that inflation negatively affects export competitiveness. Monetary policy coordination targeting price stability will structurally support export growth.
3. Improving import/export balance: The coefficient $\beta_5 = -10.74$ emphasizes the importance of industrialization policies that reduce import dependency and increase export added value.
4. Strengthening quantitative decision support infrastructure: The developed HTML-based interactive demo enables policy makers to conduct scenario analyses.

4.3 Research Suggestions

Suggestions for future research: (1) Comparison of ARIMA and VAR models at monthly frequencies with linear algebra methods. (2) Hierarchical regression models using sector-based (textile, automotive, machinery) disaggregated export data. (3) Application of Ridge and Lasso regularization methods within the SVD framework. (4) Comparison of Kernel PCA and nonlinear dimensionality reduction techniques with standard PCA.

REFERENCES

- Appetukhov, A. (2022). International trade database [Dataset]. Kaggle. <https://www.kaggle.com/datasets/appetukhov/international-trade-database>
- Awan, M. T. (2023). World export & import dataset (1989–2023) [Dataset]. Kaggle. <https://www.kaggle.com/datasets/muhammatalhaawan/world-export-and-import-dataset>
- Eckart, C., & Young, G. (1936). The approximation of one matrix by another of lower rank. *Psychometrika*, 1(3), 211–218. <https://doi.org/10.1007/BF02288367>
- Gencheva, V. (n.d.). Macro-economic indicators dataset (country-level) [Dataset]. Kaggle. <https://www.kaggle.com/datasets/veselagencheva/macro-economic-indicators-dataset-country-level>
- Goldstein, M., & Khan, M. S. (1985). Income and price effects in foreign trade. In R. W. Jones & P. B. Kenen (Eds.), *Handbook of international economics* (Vol. 2, pp. 1041–1105). Elsevier.
- Helpman, E., & Krugman, P. R. (1985). *Market structure and foreign trade: Increasing returns, imperfect competition, and the international economy*. MIT Press.
- Jolliffe, I. T. (2002). *Principal component analysis* (2nd ed.). Springer.
- Strang, G. (2006). *Linear algebra and its applications* (4th ed.). Thomson Brooks/Cole.
- Tinbergen, J. (1962). *Shaping the world economy: Suggestions for an international economic policy*. Twentieth Century Fund.